



Assessing narrow brown leaf spot severity and fungicide efficacy in rice using low altitude UAV imaging

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Highlights

- A dual-lens multispectral camera mounted on a low-altitude unmanned aerial vehicle (UAV) effectively detected narrow brown leaf spot (NBLS) in rice in the field.
- A 15-m flight altitude outperformed a 10-m flight altitude on the detection of NBLS.
- The support vector regression method was used to perform multifeature inversion modeling on NBLS.
- UAV-based images can be used to quantify the severity of the NBLS disease and evaluate the efficacy of fungicide treatments simultaneously.

Abstract

The utilization of unmanned aerial vehicle (UAV)-based imaging systems offers precise detection of plant diseases and aids in decision-making regarding fungicide applications to optimize disease control. This study employed a twin-lens multispectral camera mounted on a low-altitude UAV to capture imagery data from rice field plots that were treated with different fungicides. The objectives of this study were to assess the severity of narrow brown leaf spot (NBLS) caused by *Cercospora janseana* and to evaluate the efficacy of fungicide control. Eighteen color features and vegetation indices were extracted from RGB images, while nine color features and vegetation indices were extracted from multispectral images. Through correlation analysis, four spectral features, namely Lab-a, ExGR, VDVI, and g, were found to exhibit high correlations with disease severity. Specifically, RGB imagery had greater correlation coefficients (exceeding 0.95) for both ExGR and Lab-a features compared to multispectral imagery. A multifeature inversion modeling approach was employed, using support vector regression with the top four spectral features to predict NBLS severity. The results indicated R^2 values were above 0.93 for all support vector regressions. Furthermore, the efficacies of ten different fungicide treatments were evaluated, with UAV imaging consistently aligning with ground truth rating data in terms of efficacy ranking. These results demonstrate the potential of UAV imagery for use as a valuable tool for NBLS detection and assessing fungicide efficacy, offering significant benefits in the management of NBLS, which is a globally important disease in rice.

Introduction

Rice is one of the world's most important food crops (Jung et al., 2008). It feeds more people than any other crop and provides a major source of calories for more than half of the world's population (Kennedy and Burlingame, 2003). In the United States, rice is also a major field crop and is produced predominantly in the south-central region (Arkansas, Louisiana, Mississippi, Missouri, and Texas) and Sacramento Valley of California. However, rice farmers face challenges in managing numerous diseases that pose threats to rice production. Among these, narrow brown leaf spot (NBLS), which is caused by *Cercospora janseana*, is a global disease that can cause significant yield and quality losses, particularly in epidemic years for susceptible cultivars (Hollier, 1992; Shi et al., 2021). NBLS is more severe in late plantings and ratoon (second) crops in the southern United States, especially in years with warm springs and wet summers (Uppala and Zhou, 2019; Zhou and Jo, 2023). The pathogen infects the leaves, sheaths, and panicles of rice plants and causes lesions and discoloration on these affected tissues. Currently, fungicides remain an important tool for controlling this disease (Uppala and Zhou, 2018). The best time to apply fungicides is at the boot to heading stages (Uppala and Zhou, 2019).

Pesticides, including fungicides, are often used for rice production (Zhou et al., 2021a). However, if they are used improperly, they can lead to pollution to the environment, killing beneficial microorganisms in the soil and poisoning beneficial insects such as bees, which can damage the ecological balance of farmland. At the same time, overuse of pesticides can also lead to pesticide

resistance, leading to a gradual decrease in the effectiveness of pesticides. This will cause farmers to increase the amount and frequency of pesticides used in the control of rice diseases and other pests, which exacerbates the impact on the environment (Sah et al., 2023).

Timely and accurate detection and monitoring of crop diseases, such as NBLS, is essential for developing effective chemical control plans and evaluating the effectiveness of disease management following the use of fungicides (Ou et al., 2023). Current disease detection methods require laborious and time-consuming scouting for the presence of the target disease and assessing its severity multiple times throughout the cropping season, particularly for large acreage field crops such as rice. Moreover, the accuracy of disease assessments relies heavily on the knowledge and experience of individuals conducting the assessments, which leads to inconsistent results among different assessors (Mirik et al., 2011). Therefore, there is a growing interest in developing new methods for rapid and objective detection and assessments of crop diseases.

Traditional remote sensing applications used for pest detection, including crop diseases, rely on satellites and aircraft to collect electromagnetic radiation data in the visible, near-infrared, and far-infrared spectra at large scales (Zhang et al., 2020). However, these methods suffer from limitations in their spatial resolution and detection capabilities (Atzberger, 2013). With the advancement of multirotor systems, unmanned aerial vehicles (UAVs) have emerged as practical aerial platforms for remote sensing (Zhang and Kovacs, 2012). Compared to traditional remote sensing approaches, UAVs can operate at low altitudes and capture high-resolution images (Lan et al., 2017; Yang et al., 2022). Due to the versatility of carrying various sensors, application of UAVs in disease detection and management has gained significant attention in the field of crop protection.

For instance, Leng et al. (2012) successfully detected wheat stripe rust (*Puccinia striiformis* f. sp. *tritici*) using a UAV through linear correlation analysis between the disease index and image reflectivity. Liu et al. (2018) used UAV images that were captured at different altitudes (50, 100, 200, 300, and 400m) to detect wheat powdery mildew (*Blumeria graminis* f. sp. *tritici*). Zhang et al. (2019) proposed a method to improve the detection accuracy of wheat yellow rust (*P. striiformis* f. sp. *tritici*) by combining spectral and spatial information using multiple inception-resnet layers in a deep convolutional neural network (DCNN) based on UAV hyperspectral images and achieved an accuracy of 0.85. Zhao et al. (2019) studied the inversion of rice sheath blight (*Rhizoctonia solani* AG1-1) using a Phantom 3 Advanced equipped with multispectral sensors combined with ground truth normalized difference vegetation index (NDVI) data and disease severity ratings and achieved a prediction accuracy of up to 90.04%. These studies demonstrate the potential of UAVs in detecting and managing various crop diseases.

Various sensors, including RGB, multispectral, hyperspectral, thermal, and chlorophyll fluorescence sensors, have shown great potential for detecting crop diseases (Bock et al., 2010; Zhou et al., 2021b). Research has explored the use of RGB images to monitor crop stress based on color space and spatial information (Cogliati et al., 2015). Thermal infrared sensors have been used to detect plant diseases by capturing temperature changes related to crop water stress and microclimate

within the canopy (Jones et al., 2002; Lenthe et al., 2007). However, both RGB and thermal infrared sensors have limitations, such as a limited number of bands and susceptibility to image quality issues due to camera angle and position (Holst et al., 2020; Shrestha and Hardeberg, 2014) that somewhat restrict their effectiveness in disease detection (Mahlein, 2016). Consequently, chlorophyll fluorescence sensors have emerged as a new technology for monitoring plant photosynthesis (Meroni et al., 2009; Zarco-Tejada et al., 2009) and identifying crop diseases (Scholes and Rolfe, 2009; Tischler et al., 2018). However, chlorophyll fluorescence sensors that are used for disease monitoring require extremely high spectral resolution (<1 nm) (Damm et al., 2011). Multispectral and hyperspectral sensors that cover a broad range of wavelengths from visible to invisible light, have shown good performance in disease detection (Adao et al., 2017). Zhang et al. (2019) achieved satisfactory results using hyperspectral sensors to detect wheat yellow rust. However, hyperspectral sensors are more expensive than ordinary sensors. Yang et al. (2009) compared the detection compatibilities of multispectral and hyperspectral sensors for cotton root rot (*Phymatotrichum omnivorum*) and found that multispectral sensors were suitable for large-scale disease detection. Considering cost effectiveness and practicality, the current use of RGB and multispectral cameras is more feasible and provides effective technical means for plant disease detection.

The use of UAV platforms for rapid and automated assessments of fungicide efficacy could provide a new tool for crop protection that would reduce production costs and maximize farming returns. However, there has been limited evaluation of the potential of UAV platforms for this purpose. In a pioneering study using imagery from a ground-based digital camera, we successfully developed an image processing method that effectively and automatically assessed the efficacy of fungicide treatments for the control of wheat scab caused by *Fusarium graminearum* Schwabe under field conditions (Zhang et al., 2021). The results of this study demonstrate the feasibility of using UAV imagery to evaluate fungicide application efficacy in the field. At present, the common method for disease assessment relies on simple linear regressions of spectral information (vegetation index), but its accuracy needs improvement (Rousta et al., 2020). It would be desirable to explore the use of machine learning methods that integrate texture, spectral, and other information, including deep learning, for enhanced plant disease protection.

The objectives of the current study were to use rice NBLS, a representative foliar disease, as a model to evaluate the potential of UAV remote sensing platforms for disease detection and fungicide efficacy assessments. A multirotor UAV equipped with a multispectral sensor was deployed to capture RGB and multispectral images of rice field plots treated with different fungicides. Multiple characteristics of NBLS were analyzed and screened to develop a multicharacteristic disease grading inversion model. Based on this model, NBLS disease severities were quantified, and the efficacies of fungicide treatments were evaluated.

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Field experiment

A field experiment was conducted at the Texas A&M AgriLife Research Center (30.060298°N and 94.293370°W) in Beaumont, Texas, USA, in the 2019 crop season to evaluate the effects of 10 fungicide treatments on NBLS severity. The treatments were arranged in a randomized complete block design with four replications for a total of 40 plots (Fig. 1). The 10 fungicide treatments included: 1) untreated control; 2) Amistar Top (1.0l/ha, 18.2% azoxystrobin and 11.4% difenoconazole, Syngenta, Greensboro, ...

Changes in image characteristics as affected by fungicide treatments

Fig. 4 shows the visual results of the HSI, YCbCr, Lab color space and RGB images at 10-m and 15-m flight altitudes.

Compared to traditional RGB images, color space could provide better performance in distinguishing vegetated areas from nonvegetated areas and distinguishing damaged plant tissue from healthy areas. The visual images in all four color spaces demonstrated clear differentiation between the rice canopy and alleys (Fig. 4). However, the HSI and YCbCr color spaces exhibited greater ...

Discussion

The results of this study demonstrate the applicability of UAV imaging in detecting various levels of NBLS. In the current study, RGB images performed well for both NBLS detection and for quantifying disease severities resulting from fungicide treatments. The features extracted from the RGB images outperformed those from multispectral images in disease detection, mainly due to the ability of RGB images to more effectively capture disease symptom reflections in rice. Kaya and Gürsoy (2023) ...

Conclusion

ExGR, lab-a, VDVI, and normalized g exhibited the highest correlations with NBLS disease severity, with ExGR and lab-a having correlation coefficients of approximately 0.96. Concurrently, our assessment of the impacts of different UAV flight altitudes on the detection of NBLS indicated that a 15-m flight altitude slightly outperformed a 10-m flight altitude. The UAV disease inversion models showed varying responses to the different categories of NBLS disease severity. These results suggest that ...

CRedit authorship contribution statement

Chunyan Gu: Formal analysis, Conceptualization, Methodology, Writing – original draft. **Tao Cheng:** Visualization, Writing – original draft. **Ning Cai:** Visualization, Writing – original draft. **Weifeng Li:** Validation. **Gan Zhang:** Supervision. **Xin-Gen Zhou:** Data curation, Writing – review & editing, Supervision. **Dongyan Zhang:** Writing – review & editing, Funding acquisition, Project administration, Supervision. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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...All of the abovementioned studies have focused on constructing classification models, and with the development of unmanned aerial vehicle (UAV) remote sensing technology, many researchers have identified crop disease severity by constructing inverse models (Chen et al., 2020). For example, Gu et al. (2023) utilized a low-altitude UAV to capture RGB and multispectral images from rice fields with different severities of narrow brown leaf spot (NBLS), extracted 27 feature indices, screened four indices most relevant to rice NBLS severity by correlation analysis, and finally constructed an inversion model using these four indices and support vector regression to identify rice NBLS severity. Singh et al. (2023) used RGB and thermal infrared images combined with an inverse modeling approach to predict chickpea wilt severity....

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